

Sai Sri Harsha
SURABHI

DEVOPS ENGINEER



Passionate and aspiring DevOps Engineer with a strong foundation in Python, cloud automation, and CI/CD pipelines. Skilled in Docker and Git, with hands-on experience in deploying scalable systems and optimizing workflows. Eager to apply my skills and customer-focused insights to drive innovative solutions in a DevOps role.

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Social networks

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🦋 @SaiSriHarsha
💬 @live:surabhi.harsha

Languages

German
B2
English
C1
Telugu
Native [Mother Tongue]
Hindi
C1

Skills

AWS-DevOps | Python | Terraform | Kubernetes | Docker | Ansible | Jenkins | Gitops | Linux | BASH | PostgreSQL | MongoDB | Automation | Testing | Image Processing | PyQt | Machine Learning | Django | HTML | CSS | MATLAB | React | REST API | C++ |

Work experience

Development Engineer 06/2021 - 06/2024

Moewe Optical Solutions, Mittweida, SN, Germany

- Developed an **error detection algorithm**, utilizing image processing techniques.
- Created a cross-functional **desktop GUI** using **PyQt** to support laser functionalities.
- Updated project features with new test features following customer requirements.
- Designed function-based test scripts to ensure smooth communication processes.
- Worked with communication protocols such as I2C, UART, **TCP/IP, UDP, & Ethernet**.
- Implemented **regression testing** to enhance workflow stability and integrity.
- Developed **automated testing scripts** (pytest) to improve efficiency and accuracy.
- Conducted **unit and functional tests** to ensure system robustness.
- Developed a Petalinux environment for effective communication with FPGA.
- Created **Linux daemons** to optimize overall system functionality.
- Application containerizing using **Docker** to simplify deployment and usage.
- Utilized **git** for version control, collaboration, code management, and tracking.

Student Research Assistant (HI-Wi) 04/2020 - 09/2020

Technische Universität Chemnitz, Chemnitz, SN, Germany

- Conducted comprehensive **testing and evaluation** of 3D hand pose estimation algorithms for accuracy and performance.
- Utilized the VGG annotator to create precise ground-truth masks for hands, enhancing dataset reliability.
- **Tested the hand pose estimation** pipeline with diverse datasets, validating outputs against established ground truth values.
- **Investigated methods** to reduce message data size, contributing to more efficient data transmission.
- **Evaluated** pixel (2D) and positional (3D) **accuracy** to ensure the reliability of hand pose estimation techniques.

Software Test Engineer 10/2012 - 09/2014

Caagnota Info Pvt. Ltd., Hyderabad, India

- Created and executed comprehensive **test cases** for software function blocks, ensuring thorough **validation** of system functionalities.
- Led the execution of **Functional Testing**, System Integration Testing, and Regression Testing, contributing to overall **software quality** assurance.
- Analyzed and documented **error reports** for failed test cases, facilitating prompt resolution and continuous improvement.
- Coordinated team activities and managed project workflows, fostering effective collaboration within **Agile** frameworks.
- Worked within **Agile testing methods**, adapting to dynamic project requirements and ensuring timely deliverables.

Education

Diploma in Devops Engineer 06/2024 - 10/2024

Datascientest [Sorbonne University], Paris, France

AWS, Terraform, Kubernetes, Docker, Jenkins, Gitops, Ansible, Linux-Admin, Grafana

Integrationskurs Deutsch B1 11/2020 - 04/2021

Initiativ Gruppe, München, BY, Germany

Masters in Information and Communication Systems 10/2014 - 09/2019

TU Chemnitz, Chemnitz, SN, Germany

Image Processing, Machine Learning, Deep Learning, Computer Vision, Digital Systems

Masters in Electronics 08/2010 - 06/2012

Osmania University, Hyderabad, TG, India

Skills

Network Protocols

TCP/IP, UDP, ETHERNET

Software Programming

Python, Matlab, HTML, CSS,
C# (Basic), C/C++ (Basic)

Communication Protocols

CAN, LIN

Technical Tools

Cloud Tools

AWS Fargate, EKS, EC2,
VPC, Lambda, ECR, Gitlab,
Jenkins, Docker, Terraform,
Ansible, Grafana, ALB,
Prometheus, Datadog,
Airflow, Github

Editors and Tools

VS Code, PyCharm, Jira,
Visualstudio, Wireshark,
Trello, MicrochipStudio, H-
Term, Confluence, Arduino,
Putty

Certifications

DevOps, Ansible, Terraform,
Kubernetes, Jenkins, Docker,
SQL, NoSQL, Gitlab, Linux-
Administration, AWS
Solution Architect [Udemy],
Deutsch, ISTQB [Udemy]
[\[Link for certifications\]](#)

Other Skills

Backend Programming

- Django [RestAPI]
- FastAPI
- Flask

Manual And Autoimation Testing with Python

- GUI Tesing
- API Testing
- Manual Testing
- Electronic Board level
Testing

Operating Systems

Windows [10, 11]

Linux [Ubuntu, Fedora,
Scientific Linux]

Interests

Hiking/Trekking, Cycling.

Learning New Technologies

Projects

Devops Project [Deploying Spring-PetClinic in AWS Cloud]

06/2024 - 10/2024

Datscientest.com, Paris, France

- Deployed and managed a scalable, high-availablility infrastructure on **AWS**, leveraging services such as **EC2, S3, RDS**, Route53, and Lambda. [\[Git Link\]](#)
- Automated the provisioning of cloud resources using **Terraform** and AWS CloudFormation, significantly reduce deployment time, and streamlining setup.
- Designed and implemented **CI/CD** pipelines using **Jenkins, Gitlab CI, and AgroCD** to automate the testing, building, and, deployment of applications in AWS.
- Managed containerized applications using **Docker**, and **Kubernetes** (EKS).
- Improved application security by implemented by **IAM** roles, configuring **security groups**, and utilizing AWS Systems manager for automated security patching.
- Developed and maintained monitoring and alerting solutions using AWS CloudWatch, **Prometheus**, and **Grafana** to ensure quick incident response.
- Implemented effective **disaster recovery** solutions across multiple AWS regions, significantly reducing downtime and minimizing risks of data loss.
- Built pipelines in **Jenkins**, incorporating **Unit testing** to ensure code reliably.

Terraform [Deploying highly available application in AWS]

09/2024 - 10/2024

Datscientest.com, Paris, France

- Designed a highly available and scalable deüployment using Terraform, leveraging modular and reusable code for repeatability and efficiency. [\[Git Link\]](#)
- Configured a secure AWS environment with 1 **VPC**, multiple subnets (private and public), and NAT gateways to facilitate internet access for private resources.
- Implemented **auto-scaling groups** for EC2 instances hosting the application, to ensure performance and availability.
- Deployed **AWS RDS** across multiple availability zones to achieve database redundancy and reliability.
- Configured an Application Load Balancer (**ALB**) to evenly distribute traffic, ensure fault tolerance, and enable secure HTTPS (**TLS**) access for web servers.
- Deployed a **bastion host** in an auto-scaling group to provide a secure and scalable entry point for managing private resources.

Kubernetes Project

08/2024

Datscientest, Paris, France

- Designed and deployed Kubernetes-based microservices: FastAPI, PostgreSQL, and PGAdmin, with scalable and secure configurations. [\[Github Link\]](#)
- Deployments using YAML, **Helm** charts, and Kustomize, ensuring reusability.
- Configured HorizontalPodAutoscaler (**HPA**) to scale pods based on CPU utilization and provisioned a shared 10 Gi PersistentVolume for storage.
- Secured the application with **SSL certificates** and exposed it via **Ingress** controller.
- Prepared a **disaster recovery plan** and implemented ETCD database backups.

Instance Segmentation for Human Localization [Master Thesis]

04/2019 - 09/2019

Technische Universität Chemnitz, Chemnitz, SN, Germany

- Creating disjoint body annotations using coco-annotator for human body regions
- Detecting and localizing Humans as objects in images.
- Segmenting separate human body regions while occluding with other objects.
- Testing maximum possibility for the right regions of humans with original masks.